

PICKING THE WAVELENGTHS THAT MATTER

Autoencoder-Driven Band Selection for 4D Multi-Excitation Hyperspectral Imaging

9-13 selected (excitation, emission) bands recover full-spectrum classification accuracy on biological samples — no labels for selection

WHY

- ME-HSI = 4D cubes (x,y,λ_{ex},λ_{em})
- 200+ band combinations / pixel
- Most bands redundant or noisy
- Acquisition slow + correlated features for downstream classifiers

PROBLEM

- Existing pickers built for 3D HSI
- Variance/PCA → single bright excitation
- SPA / MCVUE → adjacent redundant bands
- None model nonlinear ex-em coupling

CONTRIBUTION (3 stages)

- 3D CAE w/ parallel excitation branches
 - Latent perturbation → influence map
 - MMR selection (relevance + diversity)
- Self-supervised. KNN validates only.

METHOD — single horizontal pipeline



Inset A — 3D CAE architecture

(parallel encoders → AVG → latent $z \in \mathbb{R}^{20}$ → shared decoder → parallel decoders)

[redraw in Publisher per spec §4D]

Acquisition + preprocessing condensed: 7-8 excitations × ~24-31 emission bands; dark / Rayleigh / mask. Not the contribution.

Inset B — Perturbation → Influence

latent dim d ●●●●●○○○○○

↓ perturb · decode

$|\Delta X(\text{ex,em})| \rightarrow \Sigma \text{ over } d$

[heatmap] [heatmap]

RESULTS — two datasets, same pipeline

Dataset A · Lichens (TPAMI submission)

sample / labels	ACCURACY ENVELOPE [must-have]	WAVELENGTH HEATMAP [must-have]
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Lichens Dataset 1 — 4 species, 192 bands. CAE selects 13 (ex,em) pairs matching baseline (86.1 % vs 85.5 % pp reduction at 0.01 pp difference by 10,000 samples).

Dataset B · Collagen Pepsin (IASIM 2026)

sample	ACCURACY ENVELOPE [must-have]	WAVELENGTH HEATMAP [must-have]
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Collagen Pepsin Dataset 1 — 100 types, 158 bands. CAE matches the 158-band KNN baseline at 5 bands; +5.8 pp at 30 bands. Selection lands on 440-490 nm = collagen autofluorescence.

MERGED RESULTS TABLE (single artifact spanning both datasets)

Configuration	Bands	Reduction	Accuracy	F1(w)	κ	Notes
LICHENS						
Baseline (full)	192	—	0.855	0.859	0.807	KNN-5 reference
Selected (best)	13	93.2 %	0.861	0.864	0.814	★ EXCEEDS baseline
Selected	9	95.3 %	0.818	0.822	0.758	
Selected	5	97.4 %	0.812	0.814	0.749	
COLLAGEN PEPSIN						
Baseline (full)	158	—	0.798	—	0.696	KNN-5 reference
Selected (best)	30	81.0 %	0.856	—	—	★ +5.8 pp
Selected	10	93.7 %	0.817	—	—	★ EXCEEDS
Selected	5	96.8 %	0.808	—	—	★ EXCEEDS

ROBUSTNESS

[robustness histogram]

Method 86.1 % vs random max 57.9 %.

The selection is not 'any 13 bands' — it's the right 13.

OBJECT-LEVEL

[per-ROI accuracy overlay]

Difficult-to-classify objects gain up to +21 pp with selected bands.

CONCLUSIONS / WHAT'S NEXT

- 90-97 % band reduction, accuracy held
 - Pipeline transfers across samples
 - Selection without supervised labels
- Next: blind drop array — pipeline recovering 3 spectral types from cubes alone.